

# Ahmed Belhareth

Senior Software Engineer | Data, AI & Agents

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## Summary

Senior Software Engineer at Linedata, specialized in Data, AI and Agents, with 3 years of experience on a business-critical financial data distribution system. Fully autonomous across the entire delivery lifecycle (dev, test, pre-production, production). Expert in Prompt Engineering and AI agent orchestration (Claude Code, Claude Opus, Kiro, MCP) for legacy system modernization and end-to-end application development (Java, Spring Boot, Angular, React).

## Technical Skills

**Languages:** Java, Python, C#, TypeScript, JavaScript, C, C++, SQL, PL/SQL, Unix/Shell (ksh, csh, sh), AWK

**Frameworks & Web:** Spring Boot, Hibernate, Maven, Angular, React JS, REST, HTML, CSS

**AI & Agents:** Claude Code, Claude Opus, Kiro (AWS Agentic IDE), MCP (Model Context Protocol), Prompt Engineering, LangChain, RAG

**DevOps & Cloud:** Docker, Jenkins, Prometheus, Grafana, AWS SageMaker, CI/CD (GitHub Actions)

**Databases:** Sybase, Oracle, MySQL, Cassandra

**Testing & Quality:** JUnit, Mockito, SonarQube, Postman

**Methodologies:** Agile, Scrum, UML

**Tools:** Git, Jira, Trello, AJIR, Genesys Cloud, IntelliJ, VS Code, Eclipse, Kiro

## Professional Experience

### Senior Software Engineer – NILE Project

09/2023 – Present

*Linedata*

- Promoted to Senior Software Engineer in recognition of autonomous ownership of critical production; annual performance review rated excellent.
- Maintenance and evolution of NILE, a business-critical financial data distribution platform (market prices, securities, issuers, MIC/CIC codes, Solvency II, NACE, LEI, Moody's and Standard & Poor's ratings) feeding the Chorus product for insurance-sector clients.
- **AI-Native Modernization:** led the migration of the NILE legacy system (UNIVAL 2.0) using Kiro (AWS Agentic IDE) and Claude Opus, reaching **12x higher velocity** than manual development (~**35 screens migrated in 3 months** vs. ~36 months with the legacy approach).
- Designed and built a **custom MCP (Model Context Protocol) server** exposing the legacy codebase context (C, Shell, PowerBuilder, Sybase stored procedures) to an AI agent, accelerating impact analysis and migration code generation toward Java/Spring Boot.
- Autonomous ownership of production management for 1.5 years: daily incident response, real-time investigation, root-cause diagnosis and fast resolution in a high-criticality environment.
- Full delivery-lifecycle administration across **four environments** (dev, test, pre-production, production): versioning, packaging and deployment via in-house Python tooling.
- Maintenance of complex batch chains orchestrated via crontab: Shell scripts (ksh, csh, sh), dynamic SQL queries, AWK processing and Sybase 15 stored procedures.
- Migrated real-time market-data distribution from C to Java, driven by Shell scripts.
- Modernization POC of the APT support interface (PowerBuilder/Sybase) to Spring Boot/Angular: designed REST endpoints consuming existing stored procedures; mentored a dedicated intern.
- Contributed to the CFT-to-MFT file transfer protocol migration alongside the infrastructure team.
- AI Ambassador: member of a 40-person Linedata program delivering weekly AI training sessions to teams.

### Final-Year Project – Web Development

02/2023 – 05/2023

- Built a web application for managing users, positions and rankings; role and permission management via Spring Security.
- Stack: Spring Boot, Angular, MySQL.

## Summer Internship – Web Development

06/2022 – 09/2022

- Built an admin dashboard for managing a restaurant's orders, menus and customers.
- Stack: Angular, Firebase (Auth, database, hosting).

## Projects

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**AfraWash – SaaS Platform (afrawash.com):** Designed and built end-to-end a car-wash PWA platform by orchestrating Claude Code: full-stack architecture, at-home and in-station booking, management dashboard, B2B marketplace, mobile conversion via Capacitor. Stack: React, Supabase, Firebase Auth.

**Image Classification:** Python, PyTorch, ResNet18 – Transfer Learning on CIFAR-10 and STL-10.

**MLOps Pipeline:** GitHub Actions, Docker – Automated CI/CD for training, testing and deploying Deep Learning models.

**NLP & RAG:** Python, LangChain – Retrieval-Augmented Generation system for document search and summarization.

**Big Data Analytics:** Spark, Cassandra, Power BI – ETL pipeline on large-scale data with interactive visualization.

## Education

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**Engineering Degree in Computer Science** **2026**

*Private College of Engineering & Technology*

**Bachelor in Computer Science – Software Engineering** **2023**

*Higher Institute of Computer Science*

**Baccalaureate in Computer Science** **2019**

*Khaznadar High School*

## Languages

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French: Fluent | English: Intermediate | Arabic: Native

## Leadership & Involvement

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**Codex Junior Enterprise:** President (2021–2022), Advisory Board Member (2022–2023)

**JCI Khaznadar:** Founding President (2017)

**Other:** VP Interact Club Atlas (2016), Member of Radio Maison de Jeunes (2018)